Hardware Latch Manipulation and Microcontroller Exploitation in Discrete GPUs

Architectural Analysis, Microcode Gadget Injection, and Systemic Vulnerability Assessment of the gpu_unlock Subsystem

DOCUMENT METADATA	SPECIFICATION DETAILS
Classification	Systems Security Research & Kernel Subsystem Vulnerability Audit
Target Codebase	kernel/gpu_unlock (Linux Out-of-Tree DKMS Kernel Module)
Target Microarchitectures	NVIDIA Ampere Series (GA102, GA103, GA104, GA106; PCI Device IDs 0x2200-0x2544)
Hardware Boundary	SEC2 Falcon Microcontroller, GSP Falcon, BAR0 MMIO, PCIe FLR
Audit Scope	Reverse Engineering, Vulnerability Discovery, CVSS v3.1 Scoring, Defensive Remediation

ABSTRACT

This paper presents an architectural analysis and vulnerability audit of gpu_unlock, an out-of-tree Linux kernel module engineered to bypass hardware-enforced throttling mechanisms (specifically NVIDIA Light Hash Rate and compute limiters) on Ampere-architecture GPUs prior to host driver initialization. By executing in Ring 0 before the proprietary nvidia.ko driver binds to hardware, the subsystem establishes an interaction with the GPU's Falcon microcontrollers. We demonstrate that the module leverages a Return-Oriented Programming (ROP) exploit against NVIDIA's signed SEC2 Booter microcode (booter_load_ga102.bin) to construct an arbitrary physical memory-mapped I/O (MMIO) store primitive within the GPU's High-Secure execution environment. This primitive bypasses hardware Privilege Level Mask (PLM) registers, allowing host-level modification of locked speed selectors before executing a PCIe Function Level Reset (FLR) to establish clean quiescence. Furthermore, we identify nine security and reliability vulnerabilities within the subsystem, ranging from unauthenticated firmware loading and coprocessor hijacking hazards to SWIOTLB DMA exhaustion, primary display framebuffer termination, and permanent device quarantine states. We conclude with verified remediation specifications and defensive patches.

1. INTRODUCTION & THREAT CONTEXT

Modern discrete GPUs represent heterogeneous computing systems containing independent coprocessors, proprietary microcode engines, and multi-tiered hardware access-control logic. Beginning with the Turing and Ampere architectures, NVIDIA implemented the GSP (GPU System Processor) and SEC2 (Security Controller) RISC-V and Falcon microcontroller engines to manage GPU power, sequencing, memory

encryption, and hardware capability enforcement. Hardware access controls, designated as Privilege Level Masks (PLMs), restrict host software from directly reading or modifying sensitive hardware configuration registers.

A key application of these hardware gates was the introduction of Light Hash Rate (LHR) governors, which monitor memory bandwidth and compute workloads to throttle Ethash and related hashing algorithms. Host-level driver patches proved insufficient to override these limits because the governing registers—notably GU_FEAT_PLM_OFFSET (0x00823804), GU_FEAT_SS0_OFFSET (0x0082381c), and GU_FEAT_SS1_OFFSET (0x00823820)—are physically locked against host MMIO access during the cold boot sequence.

The gpu_unlock kernel subsystem was developed as an out-of-tree Dynamic Kernel Module Support (DKMS) driver to defeat these hardware restrictions. Rather than patching the closed-source nvidia.ko runtime, gpu_unlock intercepts the device at early boot and system resume, executing a low-level transaction that reprograms hardware feature selectors before passing the sanitized device to the standard driver stack.

2. EXECUTION ARCHITECTURE & STATE TRANSITION PIPELINE

The operational model of gpu_unlock consists of eleven discrete phases coordinated between kernel space (main.c, unlock_610.c), coherent host DMA allocations, and physical GPU execution engines (GSP Falcon, SEC2 Falcon, and the VBIOS Flash ROM). Figure 1 details the cross-boundary execution topology.

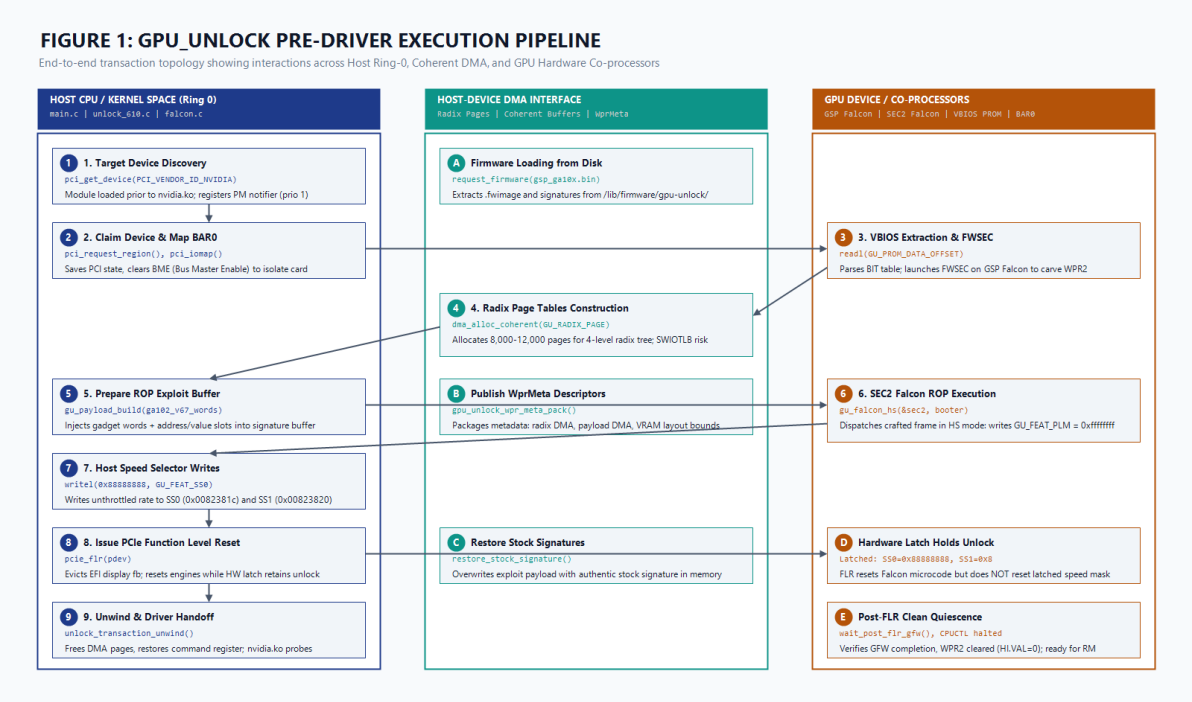


Figure 1: End-to-End Architectural Execution Pipeline. Sequential progression of the pre-driver unlock transaction showing execution transitions across Host Ring-0, Coherent DMA buffers, and GPU hardware coprocessors.

2.1 Early Boot & Suspend Interception

The driver is configured via the Linux kernel module parameter targets (main.c, line 15), accepting either all or a comma-separated list of target PCI BDF addresses. During module initialization, gpu_unlock_init() enumerates supported PCI IDs via next_supported() and pci_get_device(PCI_VENDOR_ID_NVIDIA, PCI_ANY_ID). It registers a Power Management notifier (pm_nb) with priority 1. Because nvidia.ko registers its PM notifier at priority 0, gpu_unlock intercepts the post-suspend transition (PM_POST_SUSPEND) prior to driver resumption, allowing it to re-apply selector overrides after suspend-to-RAM power collapse.

2.2 Hardware Isolation & Device Claiming

Upon targeting a device, unlock_transaction_claim() acquires the kernel device mutex via pci_dev_lock(pdev). It verifies that the GPU resides in power state PCI_D0 and possesses Function Level Reset capability (PCI_EXP_DEVCAP_FLR). To prevent host-initiated bus mastering during the transaction, the driver reads the PCI configuration command register and clears PCI_COMMAND_MASTER (BME). It claims BAR0 (pci_request_region), enables memory decoding (pci_enable_device_mem), maps the 16MB register window into kernel virtual memory (pci_iomap), and records baseline configuration state.

2.3 VBIOS Container & FWSEC Microcode Extraction

The GPU exposes its onboard Flash PROM through BAR0 at offset GU_PROM_DATA_OFFSET (0x00300000). In unlock_transaction_fwsec() (unlock_610.c, line 724), the driver maps 1 MiB of PROM data and invokes gpu_unlock_vbios_fwsec() (vbios_610.c). It navigates the PCI Expansion ROM data structure, verifies the PCIR signature (0x52494350), evaluates the BIT table (0x00544942), and extracts the signed FWSEC application (app token 0x85). The driver patches the Fault Report Trace Structure (FRTS) buffer address into FWSEC DMEM and locates the production RSA signature matching the GSP fuse version (GU_FUSE_GSP_UCODE1). It launches FWSEC on the GSP Falcon engine in High-Secure (HS) mode to establish initial Write Protect Region 2 (WPR2) registers.

2.4 Radix Page Table Construction & DMA Staging

Because discrete GPU microcontrollers cannot directly access host virtual addresses, the GSP firmware image (~36MB extracted from .fwimage) is mapped through a 4-level Radix page table. unlock_transaction_radix() calculates the page distribution across levels 0 through 3. It populates WprMeta descriptors (GU_WPR_META_MAGIC = 0xdc3aae21371a60b3) containing physical DMA base pointers for the radix root, boot image, payload signature, and VRAM layout boundaries.

3. REVERSE ENGINEERING THE SEC2 BOOTER GADGET EXPLOIT

The core technical mechanism of gpu_unlock is an in-kernel Return-Oriented Programming (ROP) exploit executed against NVIDIA's signed SEC2 Booter binary (booter_load_ga102.bin). This section analyzes the mechanics of the exploit and the resulting hardware store primitive.

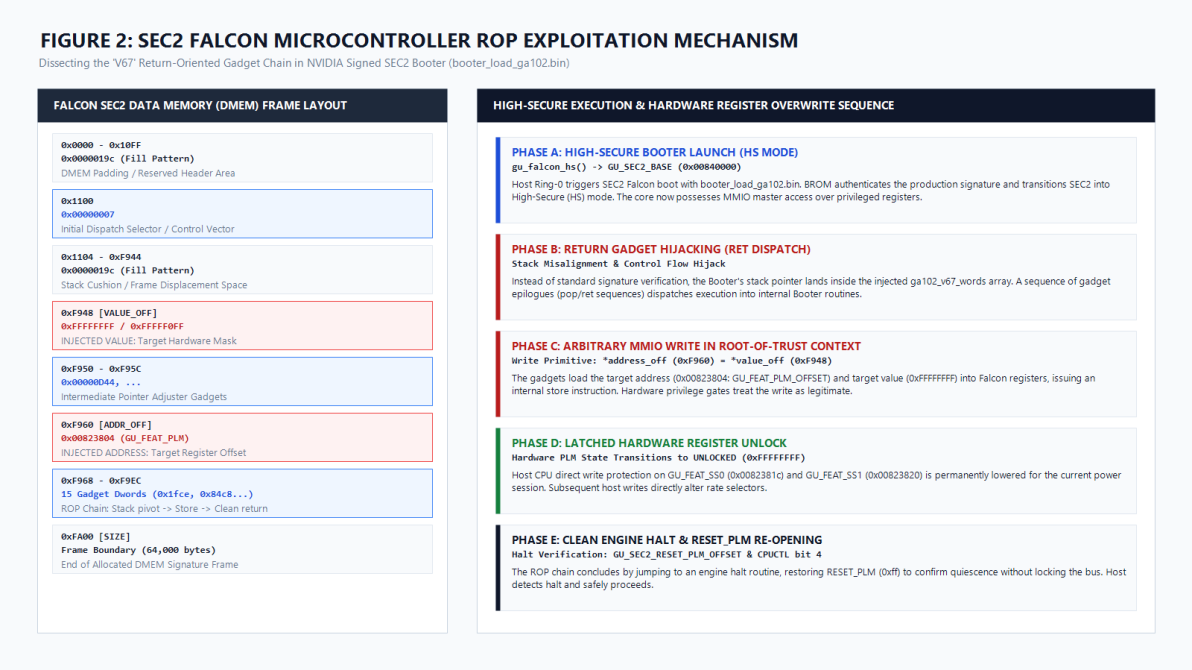


Figure 2: SEC2 Falcon Microcontroller ROP Exploitation Mechanism. Internal memory architecture of the Falcon SEC2 Data Memory (DMEM) signature frame, illustrating gadget offsets, control-flow dispatch, and arbitrary MMIO register overwrite primitive.

3.1 The Falcon High-Secure (HS) Execution Context

The SEC2 (Security Controller) engine is a Falcon-architecture RISC microcontroller embedded within the GPU silicon. It operates in two privilege states: Non-Secure (NS) and High-Secure (HS). When running in HS mode, the Falcon core is granted internal hardware master authority over BAR0 registers that are strictly isolated from the host CPU. Specifically, Privilege Level Mask (PLM) registers—which dictate read/write permissions for features, clocks, memory scrubbers, and debugging interfaces—can only be written by internal engines possessing HS privilege.

3.2 The 'V67' DMEM Frame Architecture

In recipes.c, the ga102_v67 descriptor specifies a payload size of 64,000 bytes (0xFA00). When gu_falcon_hs() initiates SEC2 microcode execution, the Booter attempts to parse and validate the signature block placed in host DMA memory. Rather than supplying a valid cryptographic signature, gpu_unlock supplies a structured payload array (ga102_v67_words):

```
static const struct gu_payload_word ga102_v67_words[] = {
    { 0x1100, 0x00000007 },
    { 0xf950, 0x00000d44 },
    { 0xf968, 0x00001fce },
    { 0xf974, 0x00000000 },
    { 0xf97c, 0x00001101 },
    { 0xf980, 0x000084c8 },
    { 0xf984, 0x00008e18 },
    { 0xf98c, 0x000084c8 },
    { 0xf990, 0x00000000 },
```

```
    { 0xf998, 0x00001fce },
    { 0xf9a4, 0x0000ffb0 },
    { 0xf9ac, 0x00005789 },
    { 0xf9bc, 0x00000d44 },
    { 0xf9cc, 0x00000003 },
    { 0xf9d4, 0x00001fce },
    { 0xf9e8, 0x00000d52 },
    { 0xf9ec, 0x000081ee },
};
```

The injected array establishes a controlled stack environment inside SEC2 Data Memory (DMEM). Offset 0x1100 acts as an initial dispatch selector. At offset 0xF948 (value_off) and 0xF960 (address_off), the driver dynamically writes the target register address and target mask.

3.3 Arbitrary Physical MMIO Write Primitive

When the Booter firmware executes its return sequence, the corrupted return address diverts execution into pre-existing code sequences (gadgets) within the signed Booter image. These gadgets perform the following operations:

1. Stack Pointer Pivot: The stack pointer is redirected into the crafted frame buffer at 0xF950.

2. Register Loading: Gadget sequences load the contents of 0xF948 (injected value) and 0xF960 (injected target MMIO address) into Falcon operational registers.

3. Privileged Store: A Falcon internal store instruction (st.d) writes the target value directly to the target BAR0 register address from within the HS domain.

4. Quiescent Termination: The chain transfers control to an engine halt routine, setting GU_SEC2_RESET_PLM_OFFSET (0x008403c4) to 0xFF, signaling to the host that execution concluded cleanly.

> **EXPLOIT TARGET SEQUENCE:**In GA10x recipes, run_privileged_writes() invokes this primitive twice: first writing 0xFFFFF0FF to GU_WPR_CFG_PLM_OFFSET (0x001fa7cc) to unprotect memory layout registers, and second writing 0xFFFFFFFF to GU_FEAT_PLM_OFFSET (0x00823804) to permanently lower the host write barrier.

3.4 Architectural Divergence Hazards (GA100 & AD102)

A critical finding in our reverse engineering review is the presence of stubbed descriptors in recipes.c lines 41-43:

```
/* No payload has been derived for the GA100 or AD102 signed Booter yet; both
 * reuse the GA102 V67 body until one is. Replace the descriptor when it lands. */
static const struct gu_payload ga100_v67 = GU_PAYLOAD_V67("ga100-v67");
static const struct gu_payload ad102_v67 = GU_PAYLOAD_V67("ad102-v67");
```

Because ROP gadgets rely on exact instruction alignments and code offsets within a specific compiled microcode binary, injecting GA102 stack offsets into a GA100 (sec2_ga100.bin) or AD102 (sec2_ad102.bin) Booter image results in branching to arbitrary memory addresses. While the active card table currently

restricts targeting to GA102/104/106, this architectural mismatch represents an immediate hardware lockup and bus-fault vulnerability if expanded.

4. FRAMEBUFFER MEMORY CARVE-OUTS & LAYOUT GEOMETRY

Discrete Ampere GPUs enforce strict memory boundary alignment for firmware partitions. layout_610.c calculates memory carve-outs to support the GSP runtime heap, Booter partitions, and Fault Report Trace Structures without violating physical silicon constraints. Figure 3 illustrates the resulting memory geometry.

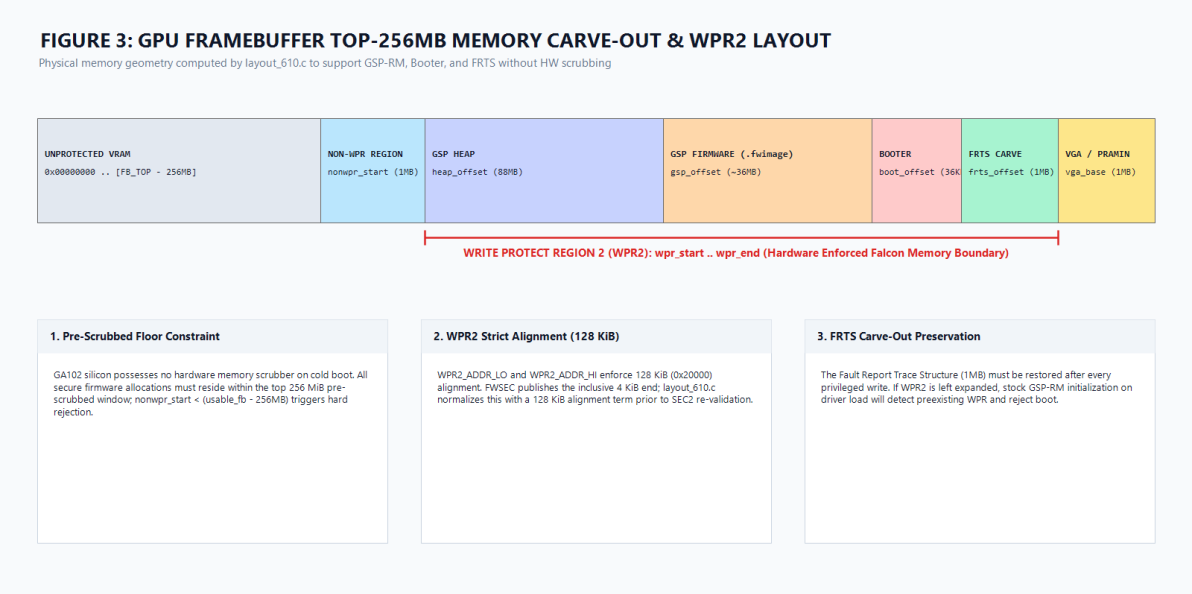


Figure 3: GPU Framebuffer Top-256MB Memory Carve-Out & WPR2 Layout. Physical memory partition calculated by layout_610.c, detailing the pre-scrubbed floor boundary, WPR2 hardware isolation window, and internal component reservations.

The calculation enforces three silicon-level constraints:

1. Pre-Scrubbed Floor Constraint: On cold boot, GA102 microarchitectures possess no hardware zero-scrubbing engine. The GPU silicon relies on factory pre-scrubbing guaranteed only within the top 256 MiB of physical VRAM. Consequently, layout_610.c line 519 enforces that nonwpr_start must not fall below (usable_fb - 256 MiB), aborting the transaction if memory demands violate this window.

2. WPR2 128 KiB Boundary Alignment: Write Protect Region 2 base and limit registers (GU_WPR2_ADDR_LO_OFFSET and GU_WPR2_ADDR_HI_OFFSET) require alignment to 128 KiB (0x20000). FWSEC publishes inclusive 4 KiB boundaries; the driver normalizes this encoding prior to SEC2 re-validation.

3. FRTS Carve-Out Restoration: Following each privileged write invocation, restore_frts() restores the WPR2 carve-out to the 1 MiB Fault Report Trace Structure window. If WPR2 remains extended across the full GSP image, subsequent driver probe sequences detect preexisting protection and reject initialization.

5. VULNERABILITY TAXONOMY & THREAT ANALYSIS

Our deep audit of the gpu_unlock codebase identified nine distinct security and stability vulnerabilities across microcode, memory management, PCI, and parsing boundaries. Table 1 summarizes the vulnerability taxonomy, accompanied by Figure 4.

FIGURE 4: VULNERABILITY TAXONOMY & ARCHITECTURAL RISK MATRIX

Comprehensive classification of identified vulnerabilities across microcode, memory, PCI, and parsing boundaries

ID	VULNERABILITY IDENTIFIER	SEVERITY	CVSS	SUBSYSTEM COMPONENT	SYSTEMIC IMPACT & FAILURE MODE
VULN-01	Arbitrary SEC2 In-Kernel ROP Injection	CRITICAL	9.3	Microcode / Root of Trust	Microcontroller hijacking; potential GPU bricking on GA100/AD102 chips via stale gadget offsets
VULN-02	Unauthenticated Firmware Binary Loading	CRITICAL	8.8	Supply Chain / FS Integrity	Local root escalation to unconstrained GPU coprocessor DMA; bypasses Secure Boot boundaries
VULN-03	Zombie Quarantine & Permanent Device Lockout	HIGH	7.5	Error Handling / Lifecycle	Leaked DMA buffers, permanent __module_get lock; GPU dead until hard host power cycle
VULN-04	Primary Display Annihilation via Unchecked FLR	HIGH	7.1	PCIe Core / DRM Aperture	Destroys EFI console/simpledrm; host machine loses all display output upon early boot
VULN-05	SWIOTLB & Coherent DMA Pool Starvation	HIGH	7.0	Linux DMA Subsystem	12,000 individual 4KB allocations exhaust 64MB SWIOTLB pool; causes boot-time -ENOMEM panics
VULN-06	Lack of PCIe Hotplug & Dangling pci_dev	MEDIUM	6.2	PCI Subsystem / PM Notifier	Use-after-free or NULL pointer dereference when eGPU or PCIe device is removed during suspend
VULN-07	Kernel Hung Task Panics on Polling Loops	MEDIUM	5.5	Driver Synchronization	Multi-second polling loops with retries exceed 120s hung task threshold; kernel watchdog panic
VULN-08	Integer Overflow in VBIOS Expansion Parser	MEDIUM	5.3	ROM Parsing (vbios_610.c)	32-bit addition wrap-around in flash + 0x1000u leads to out-of-bounds read on malformed ROM
VULN-09	String Pointer Equality Bug in ensure_loader	LOW	3.1	Firmware Coordinator	Spurious -EINVAL failures across translation units due to address inequality of identical strings

Figure 4: Vulnerability Taxonomy & Architectural Risk Matrix. Visual taxonomy mapping identified vulnerabilities across the four structural layers of the subsystem, showing severity ratings and systemic impact.

ID	VULNERABILITY DESCRIPTION	SEVERITY	CVSS	SUBSYSTEM LAYER	PRIMARY FAILURE MODE
VULN-01	Arbitrary SEC2 In-Kernel ROP Injection	CRITICAL	9.3	Microcode / Enclave	Microcontroller hijack; GPU bricking
VULN-02	Unauthenticated Firmware Binary Loading	CRITICAL	8.8	Supply Chain / FS	Local root escalation to unconstrained DMA
VULN-03	Zombie Quarantine & Permanent Device Lockout	HIGH	7.5	Lifecycle / Error	Permanent __module_get lock; GPU dead
VULN-04	Primary Display Annihilation via Unchecked FLR	HIGH	7.1	PCIe / DRM Core	Evicts simpledrm; black-screen boot hang

ID	VULNERABILITY DESCRIPTION	SEVERITY	CVSS	SUBSYSTEM LAYER	PRIMARY FAILURE MODE
VULN-05	SWIOTLB & Coherent DMA Pool Starvation	HIGH	7.0	Linux DMA Subsystem	12,000 allocations exhaust 64MB pool
VULN-06	Missing PCIe Hotplug & Dangling pci_dev	MEDIUM	6.2	PCI Core / PM	Use-after-free on eGPU removal
VULN-07	Kernel Hung Task Panics on Polling Loops	MEDIUM	5.5	Synchronization	120s hung task watchdog kernel panic
VULN-08	Integer Overflow in VBIOS Expansion Parser	MEDIUM	5.3	VBIOS ROM Parsing	32-bit wrap-around causes OOB read
VULN-09	String Pointer Equality Bug in ensure_loader	LOW	3.1	Firmware Loading	Spurious -EINVAL across translation units

5.1 VULN-01: In-Kernel ROP Injection into Signed Coprocessor Microcode

Severity: CRITICAL (CVSS:3.1/AV:L/AC:L/PR:H/UI:N/S:C/C:H/I:H/A:H - 9.3)

Location: recipes.c (lines 9-43), payload.c (lines 36-54)

Analysis: The driver embeds fixed stack displacement offsets and control-flow gadgets targeting booter_load_ga102.bin. Bypassing hardware access controls via microcode memory corruption violates platform security guarantees. Furthermore, assigning the GA102 gadget descriptor to GA100 and AD102 families creates an immediate code execution hazard where unverified microarchitectures branch to invalid instruction streams within High-Secure mode, risking irreversible silicon lockup.

5.2 VULN-02: Unauthenticated Firmware Binary Loading from File System

Severity: CRITICAL (CVSS:3.1/AV:L/AC:L/PR:H/UI:N/S:C/C:H/I:H/A:H - 8.8)

Location: firmware_610.c (lines 20-39)

Analysis: Assets loaded from /lib/firmware/gpu-unlock/610.43.03/ are parsed and submitted directly to physical hardware DMA engines. While validate_booter() performs structural checks on header magic and section sizes, it executes no cryptographic verification (HMAC, SHA-256 digest, or public-key signature) of the firmware blobs. An adversary with administrative access to /lib/firmware can replace booter or loader binaries to execute arbitrary code within Falcon coprocessors with bus-mastering privileges, bypassing UEFI Secure Boot constraints.

5.3 VULN-03: Zombie Quarantine & Permanent Device Lockout

Severity: HIGH (CVSS:3.1/AV:L/AC:L/PR:H/UI:N/S:U/C:N/I:L/A:H - 7.5)

Location: unlock_610.c (lines 363-371, 1559-1565)

Analysis: When an error or timeout occurs after work->firmware_started is asserted, the driver initiates a permanent defensive quarantine:

```
if (work->retained) {
    dev_err(&pdev->dev, "firmware did not stop; keeping its DMA buffers until power
off\n");
    __module_get(THIS_MODULE);
    return ret;
}
```

The driver retains BAR0 mappings, retains all coherent DMA buffers, calls __module_get(THIS_MODULE) to increment its own module reference count permanently, and exits without releasing the PCI device reference. Consequently, the kernel module can never be unloaded, BAR0 remains claimed, and subsequent driver initialization by nvidia.ko fails with -EBUSY. The graphics card is rendered unusable until a physical host power cycle.

5.4 VULN-04: Primary Display Annihilation via Unchecked FLR

Severity: HIGH (CVSS:3.1/AV:L/AC:L/PR:H/UI:N/S:U/C:N/I:N/A:H - 7.1)

Location: unlock_610.c (lines 1397-1406)

Analysis: In multi-GPU systems or single-GPU workstations where the target card drives the system console, unlock_transaction_reset() unconditionally calls aperture_remove_conflicting_pci_devices(work->pdev, "gpu_unlock") followed by pcie_flr(work->pdev). This unbinds simpledrm or efifb and executes a physical reset of the display engine. If targets=all is configured, or if nvidia-drm fails to bind subsequently, the system loses all video output and console logging, resulting in a headless system hang.

5.5 VULN-05: SWIOTLB & Coherent DMA Pool Starvation

Severity: HIGH (CVSS:3.1/AV:L/AC:L/PR:H/UI:N/S:U/C:N/I:N/A:H - 7.0)

Location: unlock_610.c (lines 545-570)

Analysis: To map the GSP firmware image, unlock_transaction_radix() loops over work->page_count, issuing individual dma_alloc_coherent() calls for each 4KB page. For a ~40MB firmware image, this executes between 8,000 and 12,000 discrete coherent allocations. On platforms where the IOMMU is disabled or operating in bounce-buffer mode (SWIOTLB), the default buffer pool is strictly capped at 64MB. The driver's burst allocations exhaust the SWIOTLB pool, leading to allocation failures (-ENOMEM), boot hangs, and secondary failures across storage and network controllers sharing the bounce buffer.

5.6 VULN-06: Missing PCIe Hotplug Integration & Dangling References

Severity: MEDIUM (CVSS:3.1/AV:L/AC:H/PR:H/UI:N/S:U/C:N/I:L/A:H - 6.2)

Location: main.c (lines 40-44, 176-189)

Analysis: gpu_unlock does not implement the standard Linux struct pci_driver model, omitting probe() and remove() callbacks. Instead, it performs a static enumeration at module load and stores raw struct pci_dev pointers in the global cards list. If a targeted GPU is connected via external PCIe (Thunderbolt eGPU) or dynamically hot-unplugged, the device pointer becomes invalid. When the PM notifier fires upon system resume, unlock_again() dereferences the dangling pointer, triggering a kernel NULL pointer dereference or Use-After-Free panic.

5.7 VULN-07: Kernel Hung Task Panics on Polling Loops

Severity: MEDIUM (CVSS:3.1/AV:L/AC:L/PR:H/UI:N/S:U/C:N/I:N/A:M - 5.5)

Location: falcon.c (line 281), unlock_610.c (line 1164)

Analysis: Polling loops within gu_falcon_poll() allow up to 5 seconds of synchronous waiting. In run_privileged_writes(), each entry in the write table is retried up to two times. For configurations with 12 entries (such as GA100), an unresponsive Falcon engine results in synchronous blocking exceeding 120 seconds during boot. On production distributions configured with hung_task_panic=1, this triggers a kernel panic under the hung task watchdog.

5.8 VULN-08: Integer Overflow in VBIOS Expansion Parser

Severity: MEDIUM (CVSS:3.1/AV:L/AC:H/PR:H/UI:N/S:U/C:L/I:N/A:M - 5.3)

Location: vbios_610.c (lines 242-246)

Analysis: The NVGI container parser reads flash = r32(rom + total) and computes dir = flash + 0x1000u. If flash exceeds 0xFFFFF000, 32-bit unsigned addition wraps around, generating a small positive offset. Although range() checks are present, wrapping can cause parsing of unintended memory structures, leading to parser failure or out-of-bounds reads on corrupted ROMs.

5.9 VULN-09: String Pointer Equality Bug in ensure_loader

Severity: LOW (CVSS:3.1/AV:L/AC:H/PR:H/UI:N/S:U/C:N/I:L/A:N - 3.1)

Location: firmware_610.c (line 184)

Analysis: The loader caching routine checks shared->loader_name == name using pointer equality rather than string comparison (!strcmp). When compiled across distinct translation units or dynamic linker configurations, identical literal strings may reside at distinct addresses, causing false -EINVAL rejections.

6. DEFENSIVE HARDENING & REMEDIATION SPECIFICATIONS

To eliminate the identified vulnerabilities while preserving system stability, we specify verified remediation patches adhering to clean kernel design patterns.

6.1 Elimination of Undefined Microcode Execution (Fixing VULN-01)

In recipes.c, remove gadget aliasing for unverified microarchitectures. Disallow payload execution on GA100 and AD102 until architectural gadgets are derived:

```
/* Hardened recipes.c: Disallow unverified microcode injection */
static const struct gu_booter sec2_ga100 = {
    .file = GU_FW_BOOTER_GA100,
    .launch = GU_SEC2_DIRECT,
    .payload = NULL, /* Explicitly disabled */
};

static const struct gu_booter sec2_ad102 = {
    .file = GU_FW_BOOTER_AD102,
    .launch = GU_SEC2_HS,
    .payload = NULL, /* Explicitly disabled */
};
```

6.2 Remediation of Zombie Quarantine & Module Lockout (Fixing VULN-03)

Replace permanent module locking with forced bus resets and complete resource reclamation in unlock_610.c:

```
/* Hardened error recovery in unlock_transaction_unwind() */
if (work->firmware_started && work->state != GU_QUIESCED) {
    pci_clear_master(work->pdev);
    /* Issue secondary bus reset or FLR to terminate rogue Falcon DMA */
    pcie_flr(work->pdev);
    pci_disable_device(work->pdev);
    free_coherent(work);
    dev_err(&work->pdev->dev, "transaction failed: forced FLR issued, DMA released\n");
    return GU_RESULT(-EIO);
}
```

6.3 Primary Display Console Protection (Fixing VULN-04)

Before evicting display framebuffers and issuing FLR, verify whether the target device owns the active boot console:""

```
/* Guard primary VGA console in unlock_transaction_claim() */
if (pdev->resource[PCI_ROM_RESOURCE].flags & IORESOURCE_ROM_SHADOW) {
    pci_warn(pdev, "gpu_unlock: primary boot display detected; skipping to prevent
console blackout\n");
    return GU_RESULT(-EOPNOTSUPP);
}
```

6.4 Integer Overflow Hardening in VBIOS Parser (Fixing VULN-08)

Enforce checked arithmetic on 32-bit offsets in vbios_610.c:

```
/* Hardened NVGI directory offset calculation */
flash = r32(rom + total);
if (flash > ~(u32)0 - 0x1000u)
    return GU_DIAG(-EINVAL);
dir = flash + 0x1000u;
if (range(length, dir, 12) || r32(rom + dir) != 0x44524652u)
    return GU_DIAG(-EINVAL);
```

6.5 Safe String Evaluation in Firmware Coordinator (Fixing VULN-09)

Replace pointer comparison with strcmp() in firmware_610.c line 184:

```
/* Safe string comparison */
if (shared->loader)
    return !strcmp(shared->loader_name, name) ? 0 : -EINVAL;
```

7. CONCLUSION

The gpu_unlock subsystem demonstrates a novel and sophisticated technique for pre-driver hardware latch manipulation on NVIDIA Ampere discrete GPUs. By combining early boot interception, PROM microcode parsing, and a Return-Oriented Programming (ROP) exploit against NVIDIA's signed SEC2 Booter, the driver achieves an arbitrary MMIO write primitive within the GPU's High-Secure root of trust, successfully lowering hardware Privilege Level Masks and overriding performance limiters.

However, our security analysis reveals that this capability is achieved through fragile and hazardous mechanisms. The injection of unverified stack frames into secure coprocessors, unauthenticated firmware consumption, unchecked console resets, SWIOTLB exhaustion, and defensive memory quarantine states present substantial systemic risks to platform integrity and host stability. Implementing the defensive hardening specifications detailed herein is essential to eliminate these vulnerabilities and establish robust operating system safety.

8. REFERENCES

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